

- (i) Calculate the frequency, in GHz, of the microwaves.

frequency = GHz [2]

- (ii) Show that the difference between the path lengths XQ and YQ is 6 cm.

[1]

- (iii) State and explain what may be deduced about the intensity of the microwaves at point Q.

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 [3]

- (iv) A microwave detector is positioned at P and connected to a cathode-ray oscilloscope (CRO). The controls of the CRO are adjusted so that a waveform is shown on the screen.

Describe the changes to the amplitude of the waveform as the detector is moved from P to Q.

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 [2]

[Total: 12]

